sblr Notes

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These notes describe the statistical models, data representations, sparse-LD workflows, annotation-informed priors, and technical details behind `sblr`. They are organized as a readable main path supported by practical workflow notes and more detailed technical appendices.

`sblr` is an experimental research package. The notes distinguish current public workflows from lower-level or research-stage model components and identify the assumptions needed to interpret their outputs.

1 Recommended Reading Order

New readers can follow this sequence:

1. Model Overview for the package landscape and model selection map.
2. ST-BLR Interface and Output Map for the current `stblr_csr()`, `stblr_bed()`, and `stblr_csr_annot()` interfaces, shared `Glist` object, standardized outputs, helpers, and unsupported combinations.
3. Data Representations for the relationship between individual-level data and summary statistics plus LD.
4. Single-Trait Models for the core sparse BLR model used by the current BED and CSR workflows.
5. Sparse-LD and BED Workflows for the practical path from PLINK BED data to sufficient statistics and disk-backed CSR LD.
6. Annotation-Informed Models for fixed, learned, group, and SBayesRC-style annotation priors.
7. selection_s Implementation Status for the current fixed global and sampled trait-specific BayesS-style MAF-scaling support.

8. Workflows and Outputs for running the main workflows and interpreting returned fit objects.
9. Multi-Trait Models for joint trait-sharing and covariance models.
10. Consult the technical appendices as needed for derivations, implementation contracts, and detailed assumptions.

2 Main Explanatory Notes

- Model Overview maps the main package axes: single-trait versus multi-trait, individual-level versus summary-statistics, and plain versus annotation-informed priors.
- ST-BLR Interface and Output Map maps `stblr_csr()`, `stblr_bed()`, and `stblr_csr_annot()`, the shared `Glist` object, summary-statistics and sparse-LD preparation, current model-family support, standardized outputs, helper functions, and fine-mapping diagnostics.
- Data Representations explains individual-level genotype and phenotype inputs, sufficient statistics, LD, CSR storage, and marker-order alignment.
- Single-Trait Models defines the core ST-BLR model, BayesC-style sparse prior, BED and CSR fitting paths, scheduled sparse updates, and posterior outputs.
- Multi-Trait Models introduces joint MT-BLR, trait-sharing patterns, genetic and residual covariance, sample-overlap requirements, and the current implementation status.
- Annotation-Informed Models compares fixed marker-specific priors, learned annotation effects, group priors, and the SBayesRC-style annotation-dependent BayesR prior.
- selection_s Implementation Status summarizes the current fixed global and sampled trait-specific `selection_s` support matrix, scale convention, defaults, and BayesR/SBayesRC component naming.

3 Practical Workflow Notes

- Sparse-LD and BED Workflows is the practical guide to aligning BED markers, deriving sufficient statistics, constructing sparse LD with `make_sparseLD()`, fitting `stblr_csr()`, and cautiously comparing BED and CSR fits.
- Workflows and Outputs connects the executable workflow scripts to fit-object components, compact summaries, diagnostics, interpretation, and recommended reporting.

The workflow scripts are runnable demonstrations and validation templates, not automatically lightweight package examples. Production analyses require appropriate MCMC settings, convergence checks, and sensitivity analyses.

4 Technical Appendices

- Summary Statistics derives the cross-product representation used by single- and multi-trait BLR and explains equivalence assumptions and practical limitations.
- Sparse LD and CSR defines compressed sparse row storage, the disk-backed LD contract, indexing and diagonal conventions, construction settings, and validation checks.
- Annotation-Informed Priors develops marker-specific inclusion and variance priors, empirical-Bayes constructions, learned effects, group priors, and regularization choices.
- SBayesRC-Style Annotation Priors explains the annotation-dependent BayesR mixture prior, probit stick-breaking probabilities, alpha updates, diagnostics, and differences from published GCTB SBayesRC.
- Multi-Trait Sample Overlap gives the detailed treatment of `SSY`, pairwise sample overlap, covariance reconstruction, current implementation constraints, and sensitivity analysis.

Some appendices may still be expanded as the package API stabilizes.

5 Draft Archive

The archived source drafts preserve the original long-form theory and annotation-prior material used to build the maintained notes. They remain available for provenance and research context, but the maintained notes above are the recommended reading path.

- Archived sblr theory draft
- Archived annotation-informed priors draft